Project Design Phase-II

Technology Stack (Architecture & Stack)

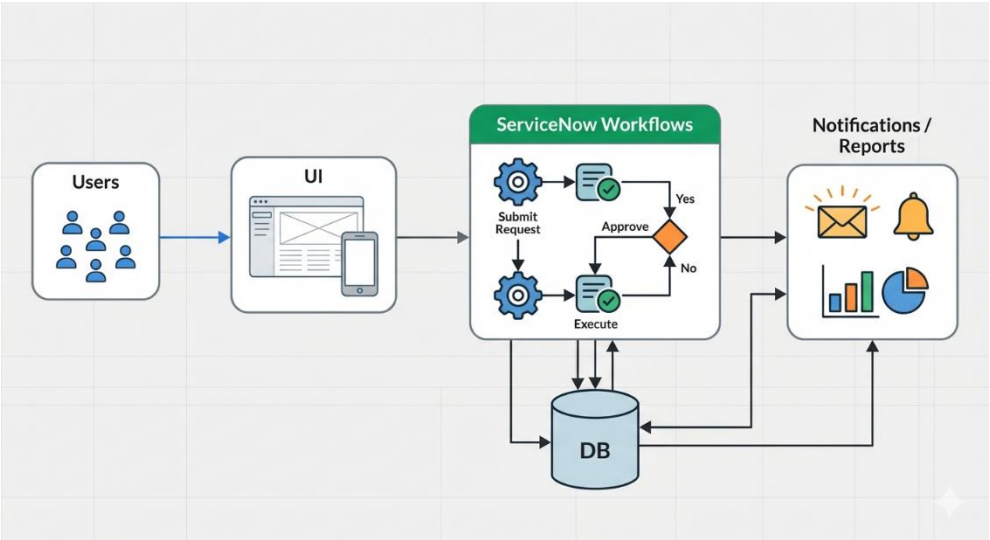
Date	13 March 2026
Team ID	69c103091466f700a6d324ef
Project Name	Smart Library Request Workflow in ServiceNow
Maximum Marks	4 Marks

Technical Architecture

The Smart Library Request Workflow is designed as a **cloud-based service management application** using **ServiceNow** as the main platform. It integrates user-facing interfaces with backend workflows, automated approvals, and database storage for tracking requests, issuance, and reporting.

Key Features of the Architecture:

- **User Layer:** Web-based interface accessible via desktop and mobile browsers.
- **Application Layer:** ServiceNow platform handles workflows, approvals, notifications, and business logic.
- **Integration Layer:** APIs for email notifications and authentication (Gmail, LinkedIn).
- **Database Layer:** ServiceNow's cloud database for storing user and request data.

- 

  Reporting Layer: Dashboard and analytics for library admin.

Reference: [IBM Pandemic Order Processing Example]

Table-1: Components & Technologies

S.No	Component	Description	Technology / Tool Used
1	User Interface	How users interact with the library workflow	ServiceNow Web UI, HTML, CSS, JavaScript
2	Application Logic-1	Workflow automation for book requests	ServiceNow Flow Designer, Business Rules
3	Application Logic-2	Approval process for librarian	ServiceNow Approval Engine
4	Application Logic-3	Notifications and email triggers	ServiceNow Notification Engine, Gmail API
5	Database	Storage for user profiles, requests, book info	ServiceNow Cloud Database (MySQL variant)
6	Cloud Database	ServiceNow backend cloud DB for scalability	ServiceNow Cloud Database
7	File Storage	Store attachments like request forms	ServiceNow Attachment Storage
8	External API-1	User authentication via Gmail	Google OAuth API

9	External API-2	User authentication via LinkedIn	LinkedIn OAuth API
10	Machine Learning	Optional: Suggest books based on user history	Python ML model / ServiceNow Predictive Intelligence
11	Infrastructure	Application Deployment on cloud	ServiceNow Cloud, Multi-region deployment

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology / Tool Used
1	Open-Source Frameworks	Frontend customization using HTML, CSS, JavaScript frameworks	AngularJS / ReactJS
2	Security Implementations	User authentication, role-based access, encrypted data, secure OAuth for Gmail/LinkedIn	SHA-256, HTTPS, OAuth2, ServiceNow ACLs, OWASP Best Practices
3	Scalable Architecture	3-tier architecture with cloud backend, microservices workflow-ready	ServiceNow Cloud, Microservices-ready Flow Designer
4	Availability	Distributed servers and cloud redundancy ensure high availability	Multi-region ServiceNow Cloud, Load Balancers
5	Performance	Efficient request handling, caching, notifications optimized, minimal page load time	ServiceNow Cache, CDN for attachments, Async workflows

References:

1. C4 Model for Software Architecture
2. IBM Online Order Processing System during Pandemic
3. IBM Cloud Architecture
4. AWS Cloud Architecture

5. Technical Architecture Diagrams Guide